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S-002 LOADING CRITERIA

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S 002 LOADING CRITERIA GENERAL

1. GENERAL

1.1 This section covers loading criteria for the design of structural and civil design.

1.2 The loading criteria for structural design shall comply with the followings, unless other criteria provided by the Owner or specified in the Bid Documents.

1.3 Reinforced Concrete design shall be designed by the ultimate strength method and to be carried out in accordance with ACI 318 or the appropriate Indonesian standards and codes of practice approved by the Engineer.

Steelwork design shall be designed by the working stress method (Allowable Stress Design) and shall be carried out in accordance with AISC or the appropriate Indonesian standards and codes of practice approved by the Engineer. The allowable stresses used shall be those given in the standards for the grade of steel specified or such other grade as shall be approved by the Engineer. The design of steel connections shall be based on Friction Type connections with ASTM A325 or A490 bolts.

1.4 All Bidders' design, detailing, calculation sheets and shop drawings shall be approved by the Owner prior to be fabrication and construction works.

2. DESIGN LOAD

Design load for structural element shall be designed in accordance with the following minimum loads.

2.1 Dead Load

Dead load consists of the structure, wall, roof, ceiling, door, window, and other necessary permanent load. Unit weight of materials shall be as follows.

Item	Load (kg/m³)
1) Steel	7,850
2) Aluminium	2,700
3) Reinforced concrete	2,400
4) Plain concrete	2,200
5) Cement mortar	2,300
6) Wood	1,000
7) Asphalt	2,300
8) Soil	1,800

9) Soil (submerged)	2,000
10) Gravel	1,950
11) Brick	2,000

2.2 Live Load (LL)

2.2.1 Live load shall comprise assumed unit loading deemed sufficient to provide moving loads, portable equipments, fixtures and parts, and also thermal shrinkage and creep load and other necessary load.

2.2.2 Live load given shall not be superimposed over the area occupied by the equipment unless there is space for possible storage or traffic under the equipment. The floors shall be designed for the removal of equipment as required per the equipment removal arrangement.

2.2.3 Live load comprises unit load sufficient to design structural frame.

The followings are the minimum requirement for live load, unless otherwise indicated by the Owner.

<u>Item</u>	<u>Load (kg/m³)</u>
1) Steam Turbine operating floor	1,250
2) HRSG floor	1,000
3) Heater floor	1,000
4) Deaerator floor	500
5) Electrical equipment floor	1,000
6) Switch gear floor	1,000
7) Control room floor	1,000
8) Computer room floor	1,000
9) Battery room floor	1,000
10) Office room floor	500
11) Toilet floor	500
12) Storage room	1,000
13) Laboratory floor	800
14) Roof (No access provided except for cleaning and repair)	75
15) Roof (General access)	150
16) Warehouse	2,000
17) Stairs and walkways	500
18) Platform and grating	500
19) Valve operating platform	2,000

20)	Grand floor of other buildings	1,250
21)	Lay down area floor	2,000
22)	Surcharge adjacent to plant structure	1,250

2.3 Machine Load (ML)

2.3.1 Machine load including pipe and cable shall comprise static, dynamic and impact load.

2.3.2 Maintenance load shall comprise significant dismantle and maintenance load as determined by the Bidder, including trolley lifted load.

2.3.3 Dynamic load shall be designed taking into account of the following loads:

- 1) Vibration such as cavitations turbine runner, eccentricity of rotor shaft, accidental force, short circuiting, vacuum load and so on.
- 2) Static torsion moment at normal and abnormal operation with dynamic condition.
- 3) Shearing stress at outer face of barrel to be induced by the maximum torsion moment.

2.3.4 The following minimum impact factors shall be used unless the Bidder uses larger values:

1)	Motors	20%
2)	Trolley loads	25%
3)	Rotating and reciprocating equipment	50%

2.3.5 Piping loads shall comprise self-weight and impact loads under operating condition. Piping expansion loads and dynamic loads will be considered on an individual design basis. Cable load shall comprise self-weight.

2.3.6 GT foundation (not applicable) and ST foundation (pedestal) shall be considered the following loads:

For GT foundation (not applicable) and ST foundation (pedestal), the following loading criteria should be considered:

- 1) Vacuum Load, 2) Normal or Operating Torque Load, 3) Thermal, Shrinkage and Creep Load, 4) Normal Machine Unbalance Load, 5) Emergency Torque Load

The details for the above loading are described in the following:

- 1) Vacuum Load

- a. The vacuum load occurs in installations which have an expansion joint between the turbine housing and the condenser. The vacuum within the condenser will impose a downward force on the pedestal deck at the turbine sole plates. The magnitude of this load is either supplied by the manufacture or computed as follows:

$$\text{Vacuum Load} = AP$$

Where: A = the area of the expansion joint opening.

P = vacuum pressure (differential between condenser interior and exterior absolute pressures).

- b. The vacuum load is distributed to the sole plates along the perimeter of the opening.
- c. The vacuum load distribution is usually provided by the vendor on the machine outline drawings.
- d. The condenser support reactions during operation would be reduced by the magnitude of the vacuum load.

2) Normal or operating torque load

- a. Normal torque loads (Lt) include the generator and turbine torque load (Tg and Tt) associated with normal power generation.
- b. Normal generator torque load (Tg), if not specified by the equipment supplier, can be computed from the formula:

$$Tg = 975.6 \text{ KW} / \text{RPM}$$

Where:

Tg = Normal generator torque in kg meters.

KW = Power Output of the unit in kilowatts.

RPM = Speed of shaft (revolutions per minute).

- c. The torque, Tg, is applied in the direction of the direction of the shaft rotation and is resolved into a vertical force couple through the sole plates along longitudinal beams supporting the generator.
- d. Normal turbine torque, Tt, is applied in the opposite direction of shaft rotation and is equal to normal generator torque multiplied by the applicable factor specified by the manufacturer. This load is distributed to each turbine in accordance with the manufacture's recommendations.
- e. The turbine torque is applied to the pedestal as a force couple acting through the machine sole plates. For low pressure turbines, the torque is resolved into a vertical force couple by dividing the

torque by the center to center distance between sole plates on the longitudinal turbine deck members. For a high pressure turbine, the torque can be applied along the sole plates on the transverse beams assuming a triangular load distribution.

- 3) Thermal, shrinkage and creep load
 - a. Thermal load include the loads associated with thermal displacements of machines, connected piping and pedestal components. Concrete shrinkage and creep are similar in effect and can be treated as equivalent thermal loads.
 - b. Machine expansion loads resulting from thermal displacements of machines are usually supplied by the equipment manufacturer. These loads include friction forces on sole plates and associated reaction forces on centering guides and anchor blocks. If these loads are not available from the equipment supplier, they can be estimated from the summation of vertical loads acting on supporting sole plates using a friction coefficient of 0.5. Vertical loads would include machine dead load, condenser loads, vacuum load, normal torque, piping reactions, etc.
 - c. Thermal loads in the pedestal result from temperature changes in the pedestal deck during machine operation and fluctuation in ambient temperatures. The global effects of these changes are accounted for by considering potential temperature differences between the mat and the deck. A minimum temperature difference of 10° should be used.
 - d. The local thermal effects associated with machine surface temperature at different locations should also be investigated. This localized heating can cause local member expansion and thermal gradients through member cross sections which can result in additional axial and flexural stresses.
 - e. Machine expansion loads and thermal loads in the pedestal are represented by T_{ol} in longitudinal direction and T_{ot} in transverse direction.
- 4) Normal machine unbalance load
 - a. Normal machine unbalance dynamic loads (LD) are vibratory loads which develop due to minor imbalance of machine rotors in vertical, longitudinal or transverse directions. These loads are sometimes specified by the machine manufacturer in terms of equivalent static loads applied at deck level. However, in most cases, they are defined as sinusoidal forcing functions acting at the center of gravity of each rotor and in a plane perpendicular to the axis of rotation. The forcing frequency and amplitude are defined in terms of rotor mass, rotational speed and a specified maxi-

imum mass eccentricity. These data (supplied or specified by the equipment manufacturer) enable definition of the forcing function as follows :

$$F = E_m \cdot M_r \cdot \omega^2$$

$$F_h(t) = E_m \cdot M_r \cdot \omega^2 \cdot \sin(\omega t + \Phi_h)$$

$$F_v(t) = E_m \cdot M_r \cdot \omega^2 \cdot \sin(\omega t + \Phi_v)$$

$$\Phi_v = \Phi_h + (\pi/2)$$

Where:

F = Magnitude of rotating force

$F_h(t)$ = Magnitude of horizontal force component at time (t)

$F_v(t)$ = Magnitude of vertical force component at time (t)

E_m = Mass eccentricity with respect to the axis of rotation (m)

M_r = Rotor mass (kg-m/sec.²)

ω = Rotor speed (radian per second)

Φ_h = Horizontal force phase angle (radian)

Φ_v = Vertical force phase angle (radian)

- b. For design purposes, the rotor speed, ω , is assumed to vary between zero and 120 percent of normal operating speed (over speed trip limit), ω_0 . The phase angle, Φ , is used to define the direction of the applied force.

- c. A representative value of E_m can be determines as follows:

$$E_m = (0.005/\omega_0)$$

Where:

ω_0 = Machine normal operating speed (radians per second)

- d. A separate forcing function is defined for each rotating machine component (e.g. generator and low, intermediate and high pressure turbine). These forcing functions are considered to act concurrently.

- e. The phase relationship between forcing functions is usually undefined or unknown and is considered to be random.
 - f. Supplemental amplified forcing functions with frequencies corresponding to the critical speeds of the rotors may also be specified by the machine supplier.
- 5) Emergency torque load
- a. Generator emergency torque can result from an accidental generator short circuit or out of phase synchronization of the generator when it is placed on line (connected to power grid).
 - b. The more severe of these two load conditions (usually the short circuit load) is specified by the equipment supplier as a time dependent forcing function or an upper bound equivalent static torque.
 - c. The generator short circuit forcing function is usually defined by two oscillating torques (one at line frequency, 50 cps, and the other at twice line frequency) superimposed on a sustained torque component.
 - d. When the load is specified as an equivalent static torque, it should be considered in either direction (clockwise and counter clockwise).
 - e. Turbine accident loads are associated with either a bowed rotor or loss of a last stage bucket.
 - f. A bowed rotor is usually postulated for high and intermediate pressure turbines. Loss of a last stage rotor blade (bucket) is usually postulated for low pressure turbines. These loads are defined in terms of time-dependent forcing functions (similar to those for machine unbalance loads) utilizing data obtained from the equipment supplier. Supplemental forcing functions corresponding to critical speeds of the rotors may also be specified by the machine supplier.
 - g. Turbine accident loads are similar to, but much higher in magnitude than, the normal machine out-of-balances loads. The accident load for each machine component is considered non-concurrent with the accident loads for other machine components.

2.4 Thermal Load (TL)

All members and elements shall be designed in consideration of load due to thermal expansion.

2.5 Construction Load (CN)

Construction load shall include significant temporary loads of false work, construction plant and equipment, etc. including any significant unbalanced loading from these items.

2.6 Crane/Hoist Load (CDL, COL and CTL)

2.6.1 The vertical and horizontal load from electric overhead traveling cranes/ monorail hoists shall be as per supplier's loading data.

2.6.2 In the absence of specific information, the following impact loads will be considered as minimum.

- 1) Vertical load at each wheel location shall be 20% of maximum wheel vertical load.
- 2) Horizontal load (longitudinal direction) at each wheel location shall be 15% of maximum wheel vertical load due to braking/traction.
- 3) Horizontal load (transverse direction) at each wheel location shall be 10% of maximum wheel vertical load.

2.6.3 Crane dead loads (CDL) shall comprise weight of trolley and bridge under a non-operating condition.

2.6.4 Crane operating loads (COL) shall comprise lift load, weight of trolley and bridge and their impact under operating condition.

2.6.5 Crane testing loads (CTL) shall comprise 125% of lift load, weight of trolley and bridge and their impact under test operating condition.

2.7 Wind Load (WL)

2.7.1 Wind load shall be calculated by multiplying velocity pressure from wind velocity on meteorological observation data by wind force coefficient from shape of structure types.

The stacks shall be designed by taking into account the oscillation due to wind pressure and the Bidder shall investigate the occurrence of resonance either as single stacks or as a group of stacks.

2.7.2 The minimum wind pressure will be calculated by use of the following equation.

$$WL = A \times P \text{ (kg)}$$

$$P = C \times q \text{ (kg/m}^2\text{)}$$

Where:

WL : Wind Load (kg)

A : Effective Area

P : Wind Pressure (kg/m²)

C : Wind Force Coefficient (refer to Table 9-1, 2)

q : Wind Velocity Pressure (kg/m²)

$$q = V^2/16(\text{kg/m}^2)$$

V : 33.3 m/sec

Table S-002-1 Wind Force Coefficient

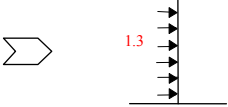
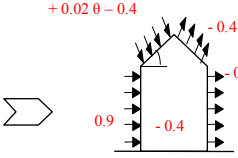
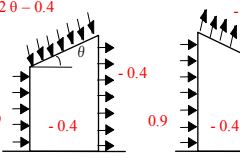
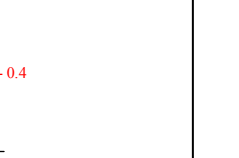
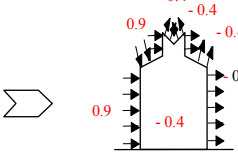
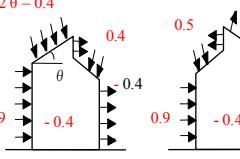
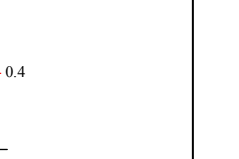
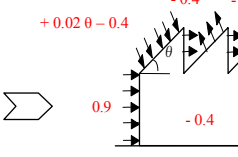
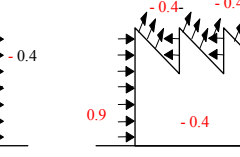
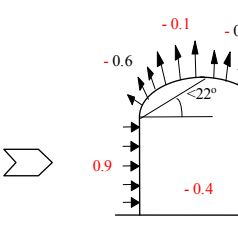
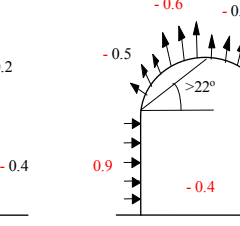
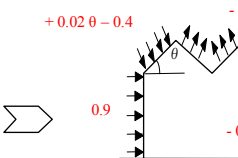
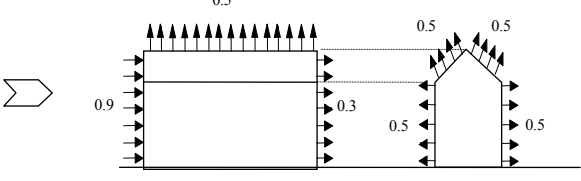
Buildings of board shape	
Buildings of closed type	           If the roof surface forms an arc, the points of inflection shall be the quarter points of the arc. In the case of wind in the ridge direction 

Table S-002-1 Wind Force Coefficient (Continued)

Buildings of open type	
Self-supporting sheds	If the slope of roof has an intermediate value between values of (a) and (b), (b) and (c) or (c) and (d), the relevant value of wind force coefficient shall be determined by linear interpolation.
Lattice structures	The below diagrams indicate cross-section of latticed beams and columns. For area affected by wind pressure, the plumb measure size of a lattice member looked from the perpendicular direction at the face of lattice shall be taken.
Net structures such as wire netting	The diagram indicates cross-section of wire netting, etc. For area affected by wind pressure, the plumb measure size of wire netting, etc. looked from the perpendicular direction at the face of wire netting, etc. shall be taken.
Circular structures such as chimneys	
In these diagrams, indicates the direction of wind, indicates the direction of wind pressure, and θ indicates the angle of roof surface with horizontal plane.	

2.8 Earthquake (Seismic) Load (EL)

2.8.1 Seismic load shall be calculated based on the latest version of Indonesian Seismic Standard i.e. Tata Cara Perencanaan Ketahanan Gempa untuk Bangunan Gedung SNI 03-1726-2012

2.8.2 Seismic load regarding the above ground parts of a building shall be calculated in accordance with the height of each part of the building. Seismic loads shall

be determined by multiplying the total dead weight and live load at each part by the seismic story shear coefficient at the height of the part.

- 2.8.3 Minimum total seismic forces used for Static Equivalent method for regular structures shall be calculated in accordance with the following formula and seismic coefficients.

$$V = C_1 \times I \times W_t / R$$

V = total static equivalent lateral force or shearing force at the base.

I = the important factor of the structure shall be determined from the table on the above seismic regulation

R = the earthquake reduction factor of the structure shall be determined from the table on the above seismic regulation

C1 = the response spectrum coefficient and shall be determined from the chart on the above seismic regulation based on soil condition for zone three.

Wt = the total dead load of the structure and vertical load with reduction factor.

For the structure that is higher than 40 m or irregular shape, the earthquake shall be determined through 3 Dimension Dynamic Response Analysis.

2.9 Soil Pressure(SP)

- 2.9.1 For sand or sandy clay

$K_a = \tan^2 (45 - \phi/2)$: Coefficient of active earth pressure.

$K_p = \tan^2 (45 + \phi/2)$: Coefficient of passive earth pressure.

$K = 1 - \sin \phi$: Coefficient of earth pressure at rest.

ϕ = Angle of internal friction

- 2.9.2 For clay, ($\phi = 0$)

(1) Active soil pressure

$$P_a = K_a \gamma h - 2C (K_a)^{0.5}$$

Passive soil pressure,

$$P_p = K_p \gamma h + 2C (K_p)^{0.5}$$

γ = Density of Soil

C = Co-efficient of Cohesion

h = Depth between ground level and level concerned
(m)

2.9.3 For all underground wall designs, a surcharge of 1250 kg/m^2 shall be applied, unless otherwise particular conditions required in the Project site.

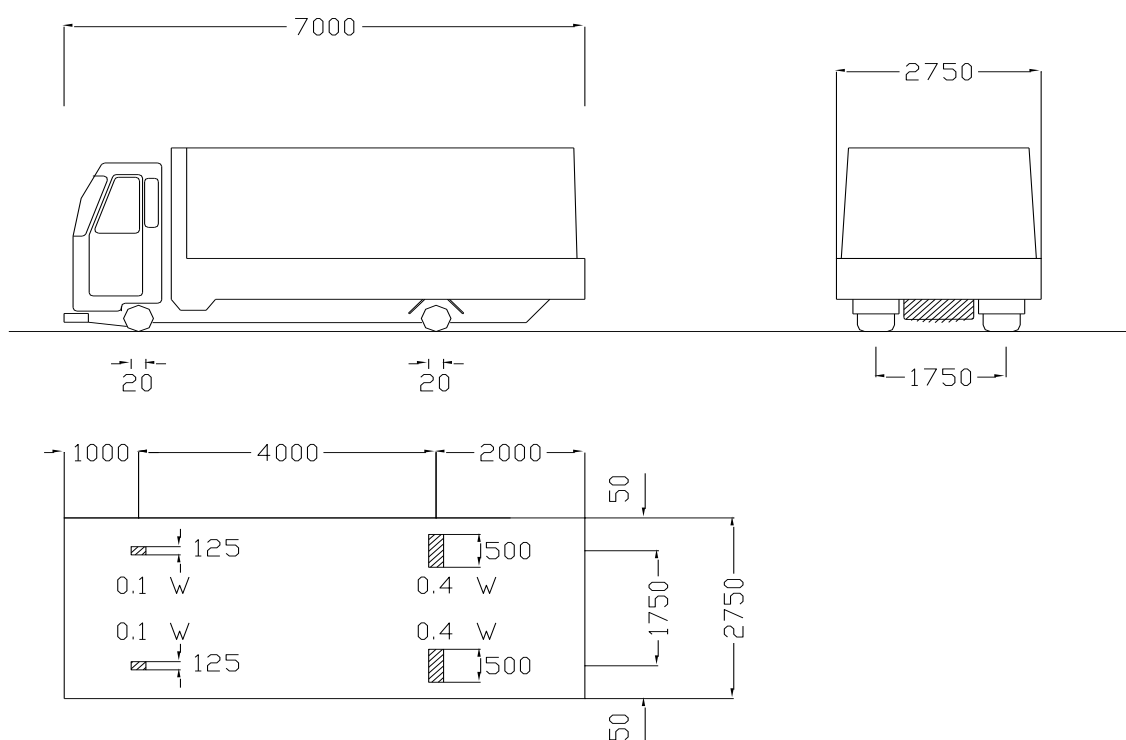
2.9.4 The water table level will be determined from the soil investigation done by the Bidder under his responsibility.

2.10 Road lane Load

Live load as to design of the road shall be used as following table:

Table S-002-3 Loading Condition for Road

Road Width (m)	Total Weight (tons)	Front wheel Load (tons)	Rear wheel Load (tons)	Width of Front wheel (cm)	Width of Rear wheel (cm)	Wheel contact length (cm)
6 and 8	25 .0	2.5	10.0	12.5	50	20
3 and 4	14.0	1.4	5.6	12.5	50	20



The minimum truck and lane loading shall conform to AASHO HS-20-44 loading.

3. LOAD COMBINATION

3.1 General

- 3.1.1 Load combination shall be calculated in accordance with the latest edition of ANSI/ASCE-7 “Minimum Design Loads for Building and Other Structures” and minimum requirement as follows.
- 3.1.2 Where a structure or member is subject to loads from two or more cranes, the crane loads shall be taken as the maximum vertical and horizontal loads acting simultaneously where this is reasonably possible.

3.2 Abbreviations of Loading

Abbreviations of loading are as follows:

Table S-002-4 Abbreviations of Loading

Loading Name		Abbreviations
Dead load		DL
Live load		LL
Roof live load		Lr
Machine load		ML
Thermal load		TL
Construction load		CL
Crane/ Hoist Load	Dead load	CDL
	Operating load	COL
	Testing load	CTL
Wind load		WL
Seismic load		EL
Soil pressure		SP

3.3 Load Combinations using Strength Design Method

Load combinations using strength design method shall be comply with the following combinations.

Table S-002-5 Load Combinations
for Reinforced Concrete Structure with Ultimate Strength Design

Case	DL	ML	CDL	LL	COL	CTL	TL	Lr	WL	EL	SP	CL	Design Strength
1	1.4	1.4	1.4										ACI 318
2	1.2	1.2	1.2	1.6			1.2	0.5			1.6		ACI 318
3	1.2	1.2	1.2	1.0				1.6					ACI 318
4	1.2	1.2	1.2					1.6	0.8				ACI 318
5	1.2	1.2	1.2	1.0				0.5	1.6				ACI 318
6	1.2	1.2	1.2	1.0						1.0			ACI 318
7	1.0	1.0	1.0						1.6		1.6		ACI 318
8	1.0	1.0	1.0							1.0	1.6		ACI 318
9	1.2	1.2		1.6	1.6		1.2	0.5			1.6		ACI 318
10	1.2	1.2		1.0	1.0			1.6					ACI 318
11	1.2	1.2						1.6	0.8				ACI 318

Case	DL	ML	CDL	LL	COL	CTL	TL	Lr	WL	EL	SP	CL	Design Strength
12	1.2	1.2		1.0	1.0			0.5	1.6				ACI 318
13	1.2	1.2		1.0	1.0					1.0			ACI 318
14	1.0	1.0							1.6		1.6		ACI 318
15	1.0	1.0								1.0	1.6		ACI 318
16	1.2	1.2		1.6		1.6	1.2	0.5			1.6		ACI 318
17	1.2	1.2		1.0		1.0		1.6					ACI 318
18	1.2	1.2		1.0		1.0		0.5	1.6				ACI 318
19	1.2	1.2		1.0		1.0				1.0			ACI 318
20	1.2	1.2	1.2	1.0					1.6		1.0	1.0	ACI 318

Table S-002-6 Load Combinations
for Steel Structure with Allowable Stress Design

Term	Case	DL	LL	ML	CDL	COL	CTL	TL	(Lr)	WL	EL	SP	CL	Design Strength
Long Term	1	1.0	1.0	1.0	1.0				1.0			1.0		1.00 AISC
	2	1.0	1.0	1.0	1.0				1.0			1.0	1.0	1.00 AISC
	3	1.0	1.0	1.0	1.0			1.0	1.0			1.0		1.00 AISC
Short Term	4	1.0	1.0	1.0		1.0			1.0			1.0		1.00 AISC
	5	1.0	1.0	1.0			1.0		1.0			1.0		1.00 AISC
	6	1.0	1.0	1.0	1.0				1.0	1.0		1.0		1.00 AISC
	7	1.0		1.0		1.0				1.0		1.0		1.00 AISC
	8	1.0	1.0	1.0		1.0			1.0	0.5		1.0		1.00 AISC
	9	1.0		1.0		1.0				0.5		1.0		1.00 AISC
	10	1.0	1.0	1.0	1.0				1.0		1.0	1.0		1.00 AISC
	11	1.0	0.5	1.0	1.0				0.5		1.0	1.0		1.00 AISC

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